databricks

Guide

The Five Pillars of Modern Analytics

Your essential compact guide to delivering fast, trusted, AI-powered analytics on the Databricks Platform

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Preface

Analytics is changing fast. Business intelligence (BI) tools and warehouses that once powered reporting now struggle with scale, real-time expectations and the rise of AI. Organizations need a new foundation: one that simplifies data architecture, unifies governance and infuses AI into the analytics workflows. This foundation must also empower every user to ask and answer questions about their data.

This compact guide outlines the **five key pillars for running modern analytics** on the Databricks Platform. Each pillar builds on the next, creating a unified, AI-ready approach to data and BI on the Databricks Data Intelligence Platform.

Pillar 1. Lakehouse Architecture: The Right Foundation for Modern Analytics

Modern analytics demands a foundation that is open, scalable and governed from end to end. The Databricks Data Intelligence Platform achieves this through lakehouse architecture that unifies the best elements of data lakes and data warehouses into a single, cohesive system. At its core, the lakehouse brings together two essential components: open storage formats on cloud object storage and centralized governance through **Unity Catalog**. Object storage becomes the system of record for analytics, while open table formats such as Delta Lake and Apache Iceberg™ provide performance, reliability, metadata management and schema evolution at scale. This keeps data flexible, portable and free from lock-in.

Unity Catalog is the governance engine that brings order and consistency to the entire analytics estate. It manages tables, schemas, files, dashboards, notebooks, AI models and pipelines in one place, with fine-grained access controls, lineage tracking, auditability and metadata enrichment. Analysts can discover assets through a unified catalog experience. Data teams gain complete observability into how data is produced and consumed. Unity Catalog also supports query federation, enabling teams to query external databases and cloud sources with consistent governance and without the need to replicate data. Quality expectations and monitoring rules can be applied directly in the catalog, ensuring that analytics consumers always work with trusted, well-understood data.

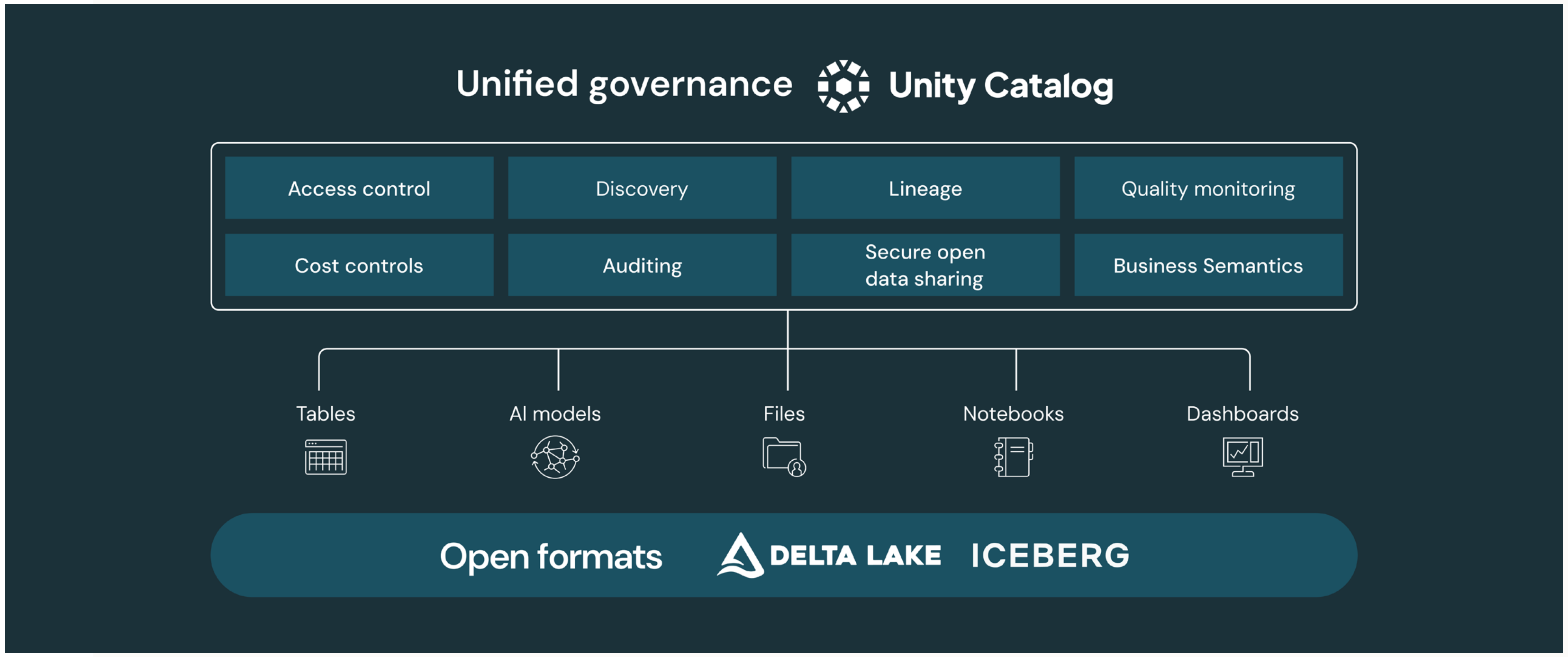

Figure 1: Lakehouse architecture combines open storage formats with unified governance

Together, open storage and unified governance create a platform that is flexible enough for data science and AI, yet disciplined enough for enterprise BI. Databricks pioneered lakehouse architecture to bring structure and governance to the scale and economics of data lakes. This design has become the modern blueprint for analytics platforms across the industry. The Databricks Data Intelligence Platform is built entirely on this lakehouse foundation, which means your BI dashboards, SQL queries, AI agents and data pipelines all operate on the same governed data in place, without extracts or duplication.

This architecture is what makes the lakehouse the strongest foundation for analytics. Dashboards and AI agents are always updated to reflect the latest data and definitions. SQL users can query local and federated sources under a single governance layer. External BI tools connect through open standards while still honoring Unity Catalog permissions. And because business semantics and metadata sit in the platform rather than in individual tools, every analytics surface delivers consistent results. The lakehouse eliminates operational friction, allowing teams to focus on insights rather than managing complexity.

Key benefits

1. **Open storage and flexibility**
Delta Lake and Iceberg on object storage ensure that analytics data is stored in open formats, giving you the freedom to work with multiple engines and tools while avoiding proprietary lock-in.
2. **Unified governance and security with Unity Catalog**
Unity Catalog centralizes permissions, lineage, quality monitoring and discovery for tables, schemas, files, dashboards, models and pipelines. Analytics teams can view the origin of data, identify who has access to it and understand its usage, all in one place.
3. **One place to access all analytics data**
Unity Catalog and Lakehouse Federation enable querying both local and remote sources while maintaining governance in a single layer. This allows BI teams to work with a comprehensive view of the business without creating fragile copies or extracts.

Learn more

- Databricks Unity Catalog web page
- Advancing the Lakehouse With Apache Iceberg v3 on Databricks
- eBook: A Comprehensive Guide to Data and AI Governance

Pillar 2. Data Engineering and Preparation: Seamless Analytics With Databricks Lakeflow

Modern analytics lives or dies on the quality and freshness of its underlying data. Yet, many teams still rely on a patchwork of ingestion tools, hand-coded batch scripts, ad hoc streaming jobs and a separate scheduler to glue everything together. This creates fragile pipelines that are difficult to debug, inconsistent to govern and slow to evolve when the business changes. The broader data engineering community has started to move toward more declarative, metadata-driven approaches, where engineers describe the desired end state of datasets and let the platform optimize scheduling, scaling and recovery.

Databricks Lakeflow incorporates this shift directly into the Data Intelligence Platform. Lakeflow is a unified data engineering solution that covers ingestion, transformation and orchestration in a single product. **Lakeflow Connect** enables point-and-click ingestion from operational databases and SaaS applications into the Databricks Platform, providing support for change data capture to keep analytics tables in sync with source systems. **Lakeflow Spark Declarative Pipelines (SDP)** lets data teams express pipelines in SQL or Python while the engine handles dependency management, incremental processing and recovery. **Lakeflow Jobs** offers workflow automation and monitoring, enabling production pipelines to be scheduled, parameterized and monitored from a single location.

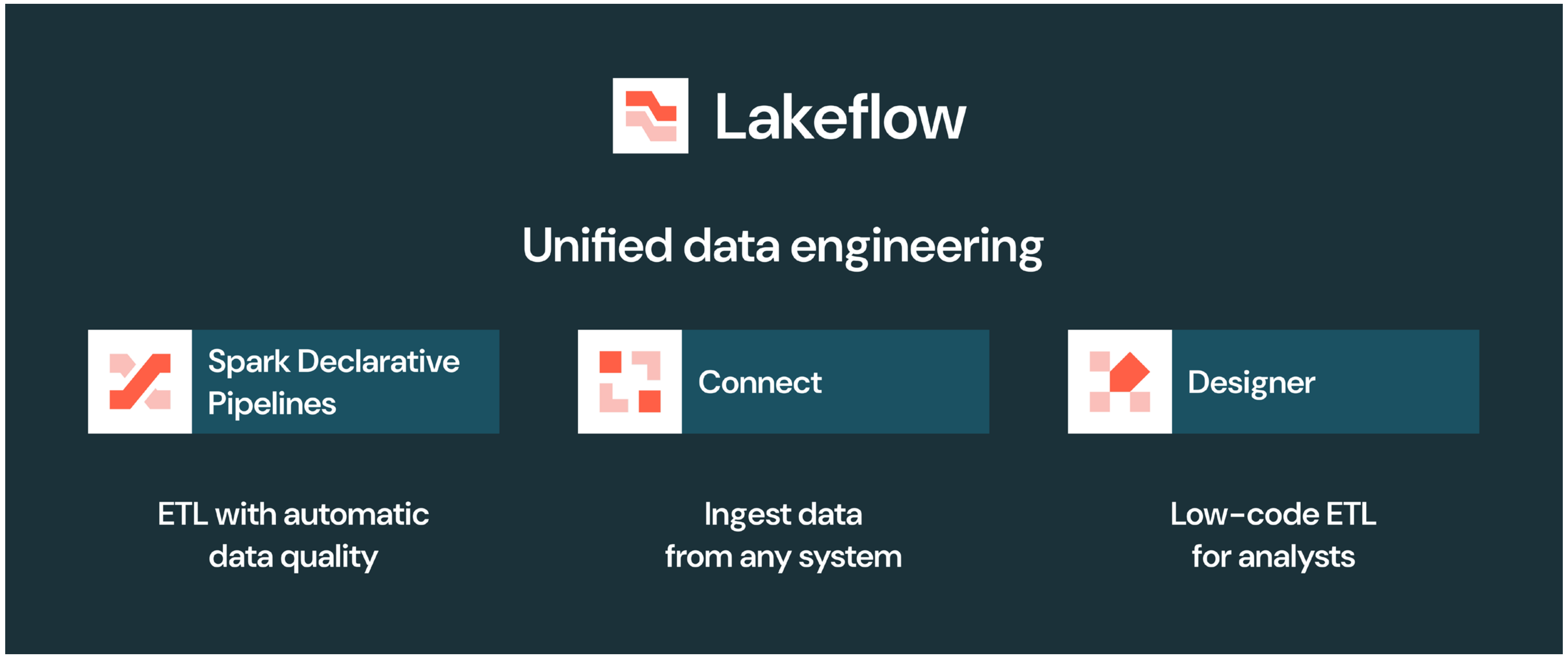

Figure 2: An overview of the products within Databricks Lakeflow

Critically, all of this is built into the same platform that runs your analytics. Pipelines run on top of the lakehouse, governed by Unity Catalog. They provide full lineage from ingestion through transformation to the tables and metric views that power dashboards and AI agents. Engineers can focus on modeling data for analytics rather than stitching together tools, while BI teams get consistent, well-described datasets with clear provenance and quality checks. External practitioners have begun highlighting Lakeflow as one of the clearest examples of this new style of declarative, metadata-centric data engineering that is tightly coupled to downstream analytics use cases.

Lakeflow also tackles a long-standing gap between central data engineering and the last mile of analytics. Many analysts still need to reshape data at the edge to finish a dashboard or answer a business question, but traditional no-code tools often create shadow pipelines that are challenging to govern. **Lakeflow Designer** addresses this directly. It's an AI-powered, no-code pipeline builder with a visual canvas and natural language interface. This lets business analysts create production-grade extract, transform, load (ETL) pipelines on the same runtime and governance layer as engineers. Visual pipelines are compiled into Spark Declarative Pipelines, eliminating the need for a separate engine and governance model and ensuring no loss of operational quality. This brings last-mile data preparation into the same lakehouse foundation, rather than pushing it into separate systems.

Because Lakeflow runs on the Databricks Data Intelligence Platform, data engineering can also leverage AI natively. Engineers and analysts can invoke AI functions within SQL pipelines to classify text, extract entities, summarize content or score sentiment as part of their transformation steps, ensuring that the datasets that feed BI already carry richer signals. Combined with the lakehouse architecture from the first pillar, this means the entire path from raw data to dashboard or agent-friendly tables is managed, observable and AI aware.

Key benefits

1. **Unified ingestion, transformation and orchestration**
   Lakeflow Connect, Spark Declarative Pipelines and Lakeflow Jobs provide a single environment for building and operating pipelines that feed analytics, from initial ingestion to incremental updates and scheduled production workflows.
2. **Declarative pipelines that are built for reliability**
   With Spark Declarative Pipelines, you describe the tables that should exist and their relationships, while the platform manages dependencies, batch or streaming modes, incremental processing and error recovery. This reduces operational toil and improves the reliability of analytics datasets.
3. **Last mile, AI-enabled data preparation for analysts**
   Lakeflow Designer provides analysts with a no-code, AI-assisted approach to refining data for dashboards and reports, while still producing production-grade pipelines that run on the same governed runtime as engineering-owned jobs. This closes the gap between central data engineering and the last mile of BI.

Learn more

- eBook: The Big Book of Data Engineering (4th Edition)
- Databricks Lakeflow product documentation
- Announcing the General Availability of Databricks Lakeflow

Pillar 3. Unified Semantics: Consistent Logic for Every Analytics Experience

As organizations adopt more tools, more personas and more AI workloads, inconsistent definitions have become one of the biggest obstacles to trustworthy analytics. Different BI tools implement their own proprietary semantic layers, each with its own modeling language and governance rules. Analysts define metrics one way in a dashboarding tool, another way in SQL scripts and a third way inside an AI prompt. As teams scale, the gap between these systems widens, leading to mismatched key performance indicators (KPIs), duplicated logic and unreliable insights. Recent industry analysis reveals that the absence of a shared semantic model is now one of the primary contributors to dashboard sprawl, poor AI accuracy and cross-team confusion.

Over the past two years, thought leaders in the analytics space have argued that the semantic layer must shift from BI tools into the data platform itself. Instead of rebuilding business logic separately in LookML, DAX or proprietary modeling languages within dashboard tools, a modern semantic layer needs to be open, SQL addressable, centrally governed and available to every downstream system. This enables BI dashboards, SQL analysts, data scientists and AI agents to all interpret business concepts through the same definitions. Databricks formalized this direction with **Unity Catalog Business Semantics**, which introduces two key components: metric views for structured, SQL-based business definitions and agent metadata for contextual knowledge that AI agents use to interpret queries. Together, they form a semantic foundation designed not only for dashboards but also for natural language interfaces and AI-driven insights.

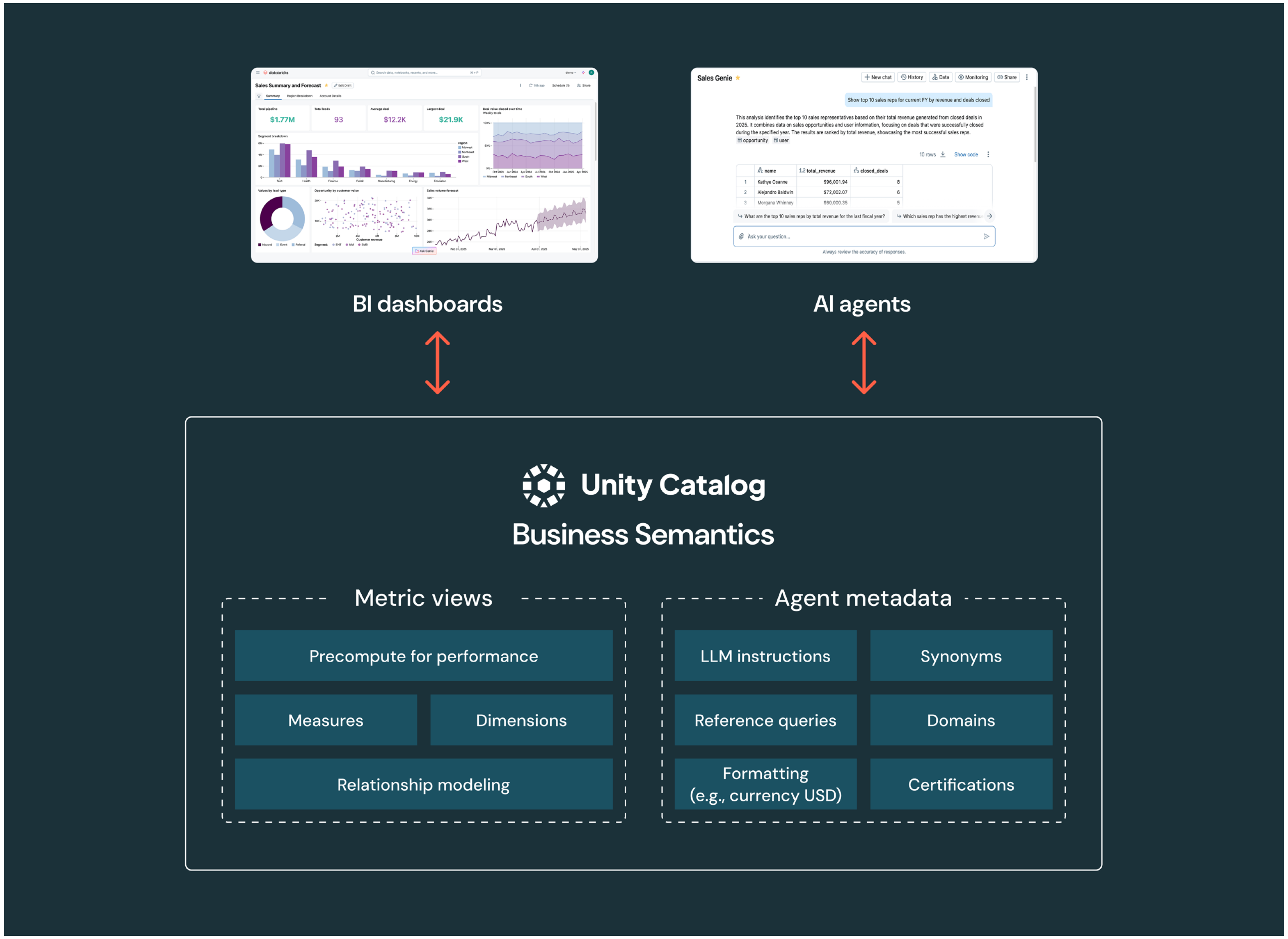

Figure 3: Unity Catalog Business Semantics combines metric views and agent metadata

Metric views provide an enterprise-wide, SQL-defined layer of business measures and dimensions. Unlike proprietary semantic layers tied to specific BI tools, metric views live inside Unity Catalog, inherit its governance and can be queried by any engine or tool that understands SQL. This creates a common foundation where dashboards, notebooks, SDP and third-party BI tools all rely on the same definitions. Organizations adopting this model report smoother collaboration between data and business teams because everyone can trace definitions back to governed, transparent logic rather than relying on black box BI metadata. The open nature of metric views also aligns with a broader industry shift toward semantic consistency as part of the lakehouse architecture highlighted in analyst research and vendor-agnostic best practices.

Agent metadata complements metric views by providing the unstructured, contextual knowledge needed for AI-driven analytics. This includes synonyms, text-based instructions, example SQL patterns and certified functions that help AI agents interpret intent. Unity Catalog serves as the storage layer for this metadata, providing agents with a comprehensive view of the data ecosystem, including lineage, usage patterns, schema details, constraints and relationships. Industry research on AI analytics reveals that agents grounded in governed enterprise metadata are significantly more reliable and produce fewer hallucinations than general-purpose large language models (LLMs).

What makes this approach especially powerful is that both metric views and agent metadata live in the same governance layer as the rest of the lakehouse. This means they benefit from Unity Catalog lineage, security, auditing, versioning and quality monitoring. Analysts no longer need to rebuild rules across multiple BI tools or train AI agents on static documents. Semantic consistency becomes part of the platform itself. As organizations adopt more AI-driven analytics, this unified semantic layer has emerged as essential — without consistent definitions across dashboards, SQL and AI agents, insight quality quickly deteriorates. With Unity Catalog Business Semantics, Databricks positions semantics not as an add-on but as a core property of the data platform.

Key benefits

1. **Consistent logic across dashboards, SQL and AI**
   Metric views define business measures and dimensions in SQL, making them accessible to any BI tool, SQL interface or AI agent without proprietary modeling languages.
2. **Semantic intelligence for AI agents**
   Agent metadata provides AI systems with the context they need to understand business terminology, query intent and dataset relevance, thereby significantly improving the quality, precision and transparency of AI-generated insights.
3. **Governance and trust, built at the platform level**
   Unity Catalog manages permissions, lineage, quality and auditing for all semantic assets. This ensures that semantic logic is transparent, traceable and consistently enforced across every analytical workflow.

Learn more

- Unity Catalog Business Semantics web page
- Unity Catalog metric views product documentation
- eBook: Redefining the Modern Semantic Layer

Pillar 4. AI-Powered Dashboards: Modern BI Built for the AI Era

Traditional dashboards were designed for a time when data volumes were smaller and analytics teams moved slowly. They often rely on complex extracts, pre-aggregations, cubes, proprietary semantic layers and separate licensing for each user. In today's fast-paced environment, where organizations expect real-time insights, flexibility and broad self-service, those legacy constraints create governance drift and limit scalability. Modern analytics requires dashboards that are native to the data platform, centrally governed and enhanced with AI.

On the Databricks Platform, **AI/BI Dashboards** meet this need directly. They run in place on your lakehouse, powered by governed data and shared semantics, eliminating data extracts and license friction entirely. Because they use Unity Catalog for access control, lineage and metadata, dashboards are always consistent with the rest of the platform. They also run on top of metric views, which provides two important advantages for BI teams. First, metric views ensure that every dashboard and user works from the same definitions, avoiding mismatched KPIs. Second, their materialized structure can significantly improve performance for common measures and aggregations.

Built-in AI assistance further accelerates the creation and consumption of visual analytics. Authors can describe a question or visual in natural language, have the platform generate SQL and chart configurations, and rapidly iterate without having to write everything by hand. Dashboards also support integrated conversational exploration through AI agents such as AI/BI Genie. This allows business users to ask follow-up questions in natural language and get accurate answers without waiting for data teams or requiring new dashboard pages. The combination of interactive visuals and conversational analytics provides users with a deeper, more intuitive understanding of their data.

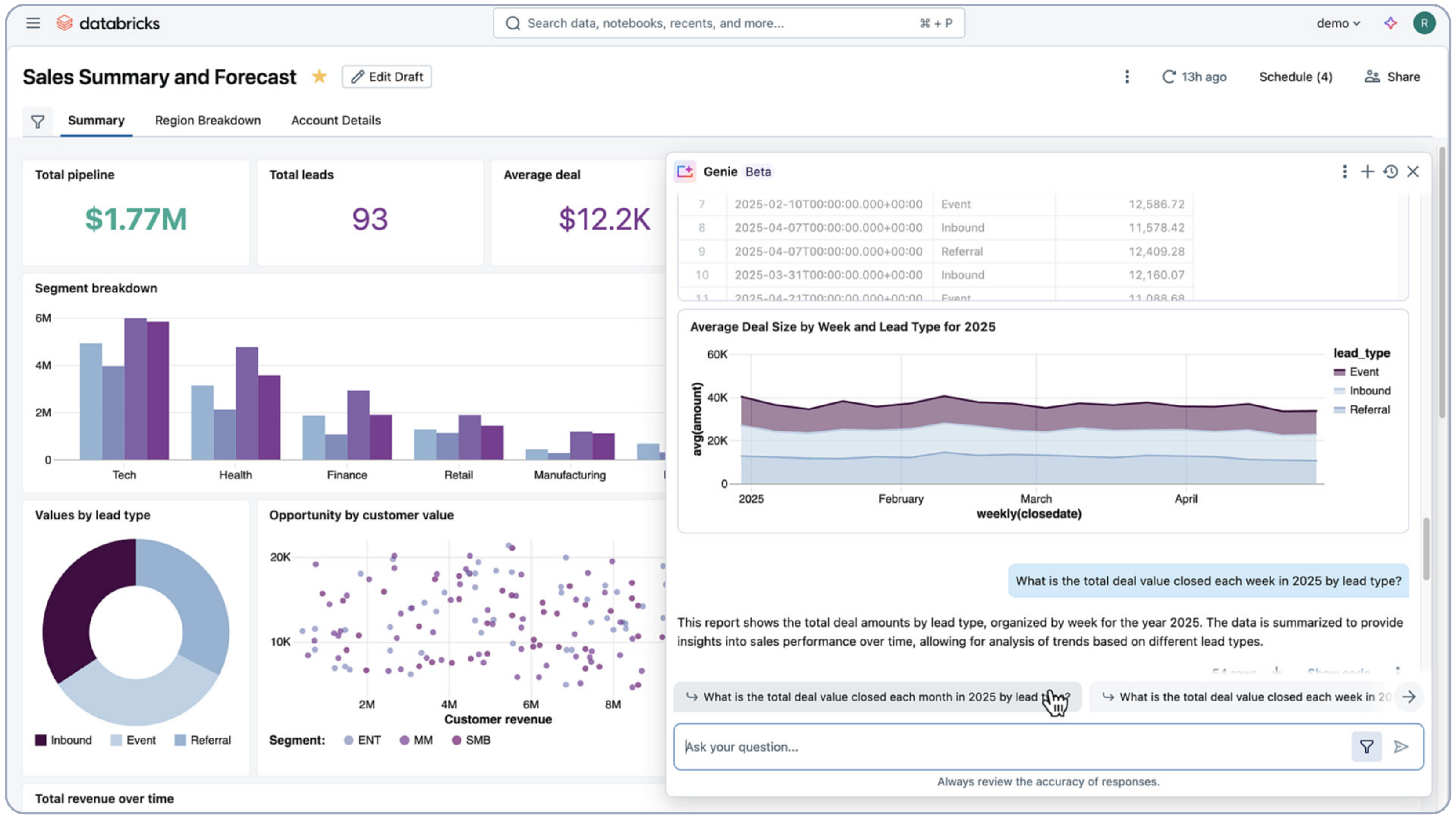

Figure 4: Screenshot of a Databricks AI/BI Dashboard with an integrated Genie

Organizations that adopt this model see immediate benefits. Dashboards become simpler to author, faster to explore and accessible to the entire business, not only technical users. Independent evaluations of AI-enhanced dashboards show that this approach strengthens self-service, reduces dependence on data teams and accelerates decision-making with real-time and predictive insights.

This shift addresses the largest pain points of traditional BI, including data extract sprawl, rigid scheduling, inconsistent semantics, licensing costs and slow turnaround for new questions. AI/BI Dashboards combine the power of the lakehouse with the flexibility of AI to deliver fast, consistent and scalable analytics across the organization.

Key benefits

1. **Native integration with the data platform**
   Because AI/BI Dashboards run directly on your lakehouse and Unity Catalog, there's no need for data duplication, separate BI servers or data extracts. Governance, semantics, lineage and access controls remain unified with the rest of your data assets.
2. **AI-driven creation and rapid dashboard development**
   Users can describe visualizations, datasets or even entire dashboards in natural language. The built-in AI assistant generates the SQL, builds visuals and delivers a working dashboard in minutes. This lowers the barrier for dashboard creation and dramatically speeds up time to insight.
3. **Self-service and interactive analytics for all users**
   Dashboards support interactive features, including filtering, drilling, cross-filtering, dynamic parameters and the embedding of AI-driven widgets (such as forecasting, anomaly detection or predictive analytics) — helping both technical and nontechnical users explore and derive insights on the fly.
4. **Cost-effective, scalable BI without per-user licensing**
   Unlike many legacy BI tools that charge per seat or per viewer, AI/BI Dashboards on Databricks only consume compute when queries run. There aren't any additional per-user license fees. This makes scaling dashboards across the organization far more economical and removes friction to wider adoption.

Learn more

- Databricks AI/BI Dashboards product page
- Databricks AI/BI Dashboards product documentation
- Tutorial: AI/BI Dashboards Embedding

Pillar 5. Conversational Agents: Self-Service Insights With Databricks AI/BI Genie

Dashboards and SQL notebooks give analysts more power, but they often leave business users on the outside. For many teams, getting a new question answered still means logging a request, waiting in a queue and receiving a static report that is already stale. Over the past couple of years, AI has begun to alter this pattern. Industry commentators now describe a shift from tool-centric analytics to conversation-centric analytics, where users simply ask questions in natural language and AI agents do the work of finding data, writing queries and explaining results.

The challenge is that generic chatbots don't understand an organization's data sufficiently to drive informed decisions. They're trained on the public internet and are unaware of a company's internal schemas, metrics or business rules. That's why there's a growing consensus that effective conversational analytics must be grounded in a governed lakehouse and a shared semantic layer, not just an LLM.

Genie is built exactly on this principle. It's a conversational analytics agent that sits directly on the Databricks lakehouse foundation and Unity Catalog, providing governed access to real tables, metric views and metadata, rather than brittle extracts. Users ask questions like "How did net revenue trend by segment this quarter?" and Genie responds with text summaries, tabular results and optional charts, along with an explanation of how it arrived at the answer.

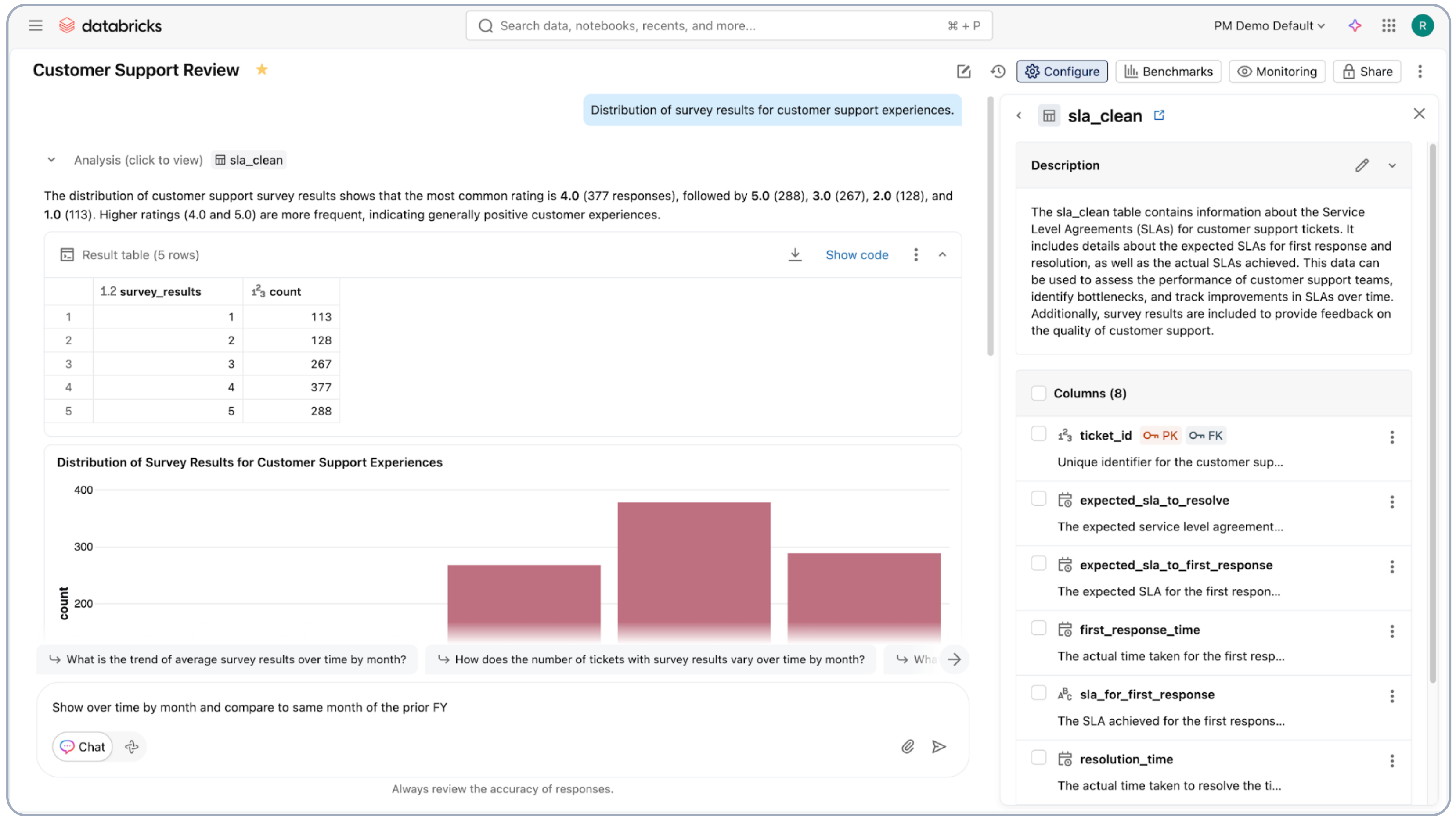

Figure 5: Screenshot of a user asking natural language questions in Databricks Genie

Crucially, Genie isn't a one-size-fits-all chatbot. Each Genie space is tuned to a particular subject area and powered by Unity Catalog Business Semantics. Metric views provide structured definitions for measures and dimensions, and Genie uses this layer when generating queries and interpreting questions. Additionally, the Genie knowledge store enables authors to curate localized context for each space. They can add table and column descriptions, synonyms, sampled values, value dictionaries and structured SQL instructions that guide Genie in answering questions.

Genie also learns continuously from how people use it. When users provide feedback with thumbs up or thumbs down, high-quality answers can be captured as reusable knowledge snippets in the knowledge store. This allows these snippets to be reused in future conversations.

Over time, a Genie space becomes a living knowledge system that combines governed data, shared semantics, curated instructions and proven answers. Practitioners who've implemented Genie at scale report that this combination of grounding, tuning and learning is what allows Genie to move from a novelty to a reliable self-service analytics channel.

Finally, AI/BI Genie and AI/BI Dashboards are exposed through **Databricks One**, so users don't need to know where a particular dataset or dashboard lives. They simply open a unified interface, choose a relevant dashboard or start asking questions, which are automatically routed to the Genie spaces they can access.

Key benefits

1. **Deeply grounded in your real data and semantics**
   Genie runs on the lakehouse with access controlled by Unity Catalog and guided by metric views. It generates SQL against governed tables instead of relying on opaque embeddings or static extracts.
2. **Rich tuning and continuous learning through the knowledge store**
   Authors can shape how Genie behaves by adding metadata, synonyms, example queries and instructions. Genie then learns from user feedback, capturing high-quality answers as reusable knowledge and improving over time.
3. **Scalable self-service in a single interface**
   Through Databricks One, any authorized user can access the relevant Genie spaces and ask questions in natural language. They'll receive trusted answers without needing to open a BI tool or write SQL. This reduces reliance on central analytics teams and speeds up decision-making.

Learn more

- Databricks AI/BI Genie web page
- Databricks AI/BI Genie product documentation
- eBook: Business Intelligence Meets AI

How to Get Started With Modern Analytics on Databricks

Modern analytics is most powerful when teams can explore, prepare, govern, visualize and question their data all in one place. With the Databricks Data Intelligence Platform, every pillar mentioned in this guide — lakehouse architecture, Lakeflow, unified semantics, AI-powered dashboards and Genie — is available the moment you start. Getting hands on is the fastest way to understand how these capabilities work together to shape a modern analytics practice.

Whether you're new to Databricks or already using the platform, the steps below will help you move from exploration to impact quickly.

For new users: Try Databricks Free Edition

If you're not yet a Databricks customer, the **Databricks Free Edition** is the easiest way to get started. It includes preloaded examples, allowing you to experience the full analytics workflow without setup.

Start here: Databricks Free Edition

With Free Edition, you get:

- Sample datasets curated for analytics scenarios
- Pre-built AI/BI Dashboards to explore interactivity and performance
- Out-of-the-box Genie spaces to test conversational analytics
- Guided notebooks for SQL, data prep and lakehouse concepts

This environment gives you a hands-on feel for the platform and lets you experience the unified workflow described throughout this guide.

For existing Databricks customers: Three ways to get started today

If your organization already uses Databricks, you can accelerate adoption of modern analytics by enabling and exploring the core capabilities referenced across the five pillars.

1. START WITH AI/BI DASHBOARDS IN YOUR WORKSPACE

Learn how to build dashboards, use the Databricks Assistant to generate SQL and visualizations and embed dashboards in applications. Refer to the AI/BI Dashboards documentation for more information.

2. CREATE YOUR FIRST GENIE SPACE

Bring conversational analytics to life in your existing data environment using Unity Catalog semantics and the Genie knowledge store. Refer to the AI/BI Genie documentation for more information. To tune Genie for a real dataset, start with the knowledge store overview.

3. BUILD YOUR FIRST GOVERNED SEMANTIC MODEL

Create metric views to define reusable measures and dimensions for dashboards, SQL users and Genie. Read the metric views documentation to learn how. For a broader understanding of business semantics, please visit the Unity Catalog Business Semantics product page.

Resources

In addition to the links already provided in this guide, the following is a curated set of blogs, product documentation, web pages, technical papers, eBooks, customer stories and third-party best practices that deepen your understanding of the five pillars and how to implement them effectively.

Databricks lakehouse architecture: Foundation for modern analytics

- Announcing full Apache Iceberg support in Databricks
- What's New With Databricks Unity Catalog at Data and AI Summit 2024
- Unity Catalog Governance in Action: Monitoring, Reporting and Lineage
- A Year of Interoperability: How Enterprises Are Scaling Governance With Unity Catalog
- Databricks Unity Catalog Implementation: A Complete Guide to Secure and Scalable Data Governance
- Unity Catalog and Open Table Formats: A Guide
- Data Governance Frameworks on Databricks: A Role for Unity Catalog
- Lakehouse format convergence: Delta Lake and Apache Iceberg
- eBook: Big Book of Data Warehousing and BI

Databricks Lakeflow: Data engineering and preparation

- Bringing Declarative Pipelines to the Apache Spark™ Open Source Project
- Lakeflow Jobs product documentation
- Lakeflow Spark Declarative Pipelines concepts
- Databricks Unveils Lakeflow Designer for Data Analysts to Build Reliable Pipelines
- No-Code Data Operations: How Business Analysts Can Build Self-Service Analytics on Databricks
- Databricks Lakeflow for Modern Data Engineering
- Declarative Pipelines: The Next Major Abstraction in Data Engineering?

Databricks Unity Catalog: Unified business semantics

- Unity Catalog Governance in Action
- What's New With Databricks Unity Catalog at Data + AI Summit 2024
- Unity Catalog Semantic Layer
- How Unity Catalog Metrics Redefines Trusted KPIs
- Cementing Your Business Semantics to Power Analytics and AI

Databricks AI/BI: AI-powered dashboards

- Databricks AI/BI product page
- What's New in AI/BI — October 2025 Roundup
- AI-Driven Business Intelligence: Smarter Dashboards, Faster Decisions
- Tutorial: AI/BI Dashboards Embedding
- How to embed AI/BI Dashboards in customer-facing applications
- Author Faster, Smarter AI/BI Dashboards With Dynamic Calculations

Databricks AI/BI Genie: Conversational agents

- AI/BI Genie is now Generally Available
- Onboarding your new AI/BI Genie
- What is an AI/BI Genie space
- Build a knowledge store for more reliable Genie spaces
- 5 key lessons from implementing AI/BI Genie for self-service marketing insights
- Databricks AI/BI Genie: The Future of Conversational Analytics
- Part 2: Diving into the Databricks Genie knowledge store
- AI/BI Genie: A compound AI system
- Your lakehouse just got a voice: How AI agents and natural language interfaces are reshaping analytics
- Conversational Analytics: The Future of Data Interaction
- How AI is changing the analytics stack

Videos, product tours and tutorials

- Analytics and BI demo library

Real-world examples

- Databricks AI/BI customer stories
- Databricks Lakeflow customer stories
- Databricks SQL customer stories
- Databricks Unity Catalog customer stories

Conclusion

Analytics is changing faster than ever and the organizations that thrive will be the ones that modernize their approach. The five pillars outlined in this guide all reinforce the same idea: Analytics become dramatically more powerful when your data, semantics, dashboards and AI systems all reside in one governed platform. With a lakehouse foundation, automated data engineering, shared business definitions, AI-assisted dashboards and conversational access through Genie, Databricks enables teams to work the way they've always wanted to: faster, simpler and with far greater impact.

With these capabilities in place, analysts spend less time preparing data and more time uncovering insight. Business users shift from waiting in queues to answering their own questions in seconds. AI becomes a trusted copilot for exploring data and refining logic rather than an isolated experiment. And leadership gains transparency, alignment and confidence across the entire analytics lifecycle.

This is the new standard for modern BI. And thanks to Databricks Free Edition and the built-in experiences across Databricks One, getting hands on has never been easier.

About Databricks

Databricks is the data and AI company. More than 20,000 organizations worldwide — including adidas, AT&T, Bayer, Block, Mastercard, Rivian, Unilever and over 60% of the Fortune 500 — rely on Databricks to build and scale data and AI apps, analytics and agents. Headquartered in San Francisco with 30+ offices around the globe, Databricks offers a unified Data Intelligence Platform that includes Agent Bricks, Lakeflow, Lakehouse, Lakebase and Unity Catalog. To learn more, follow Databricks on LinkedIn, X, YouTube and Instagram.

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